MATHEMATICS PROJECT WORK

Formative Assessment 2 (FA 2)

1. Name of the Project

Classifying the situations of finding the zeroes of a quadratic polynomial $y = ax^2 + bx + c$ by drawing its graph with appropriate explanations.

2. Objectives

- To understand how the graphs are formed.
- To find out the number of zeroes in each case.
- To recognize the shape of the graphs.

3. Data Analysis

Case (i): The graph intersects the X-axis at two distinct points.

In this case, the graph cuts the X-axis at two distinct points, say A and A'. Therefore, the x-coordinates of points A and A' are the two zeroes of the quadratic polynomial $ax^2 + bx + c$. This parabola can open either upwards or downwards.

Example: Let us find the values of x and y for the quadratic polynomial $x^2 - x - 6$.

$p(x) = x^2 - x - 6$

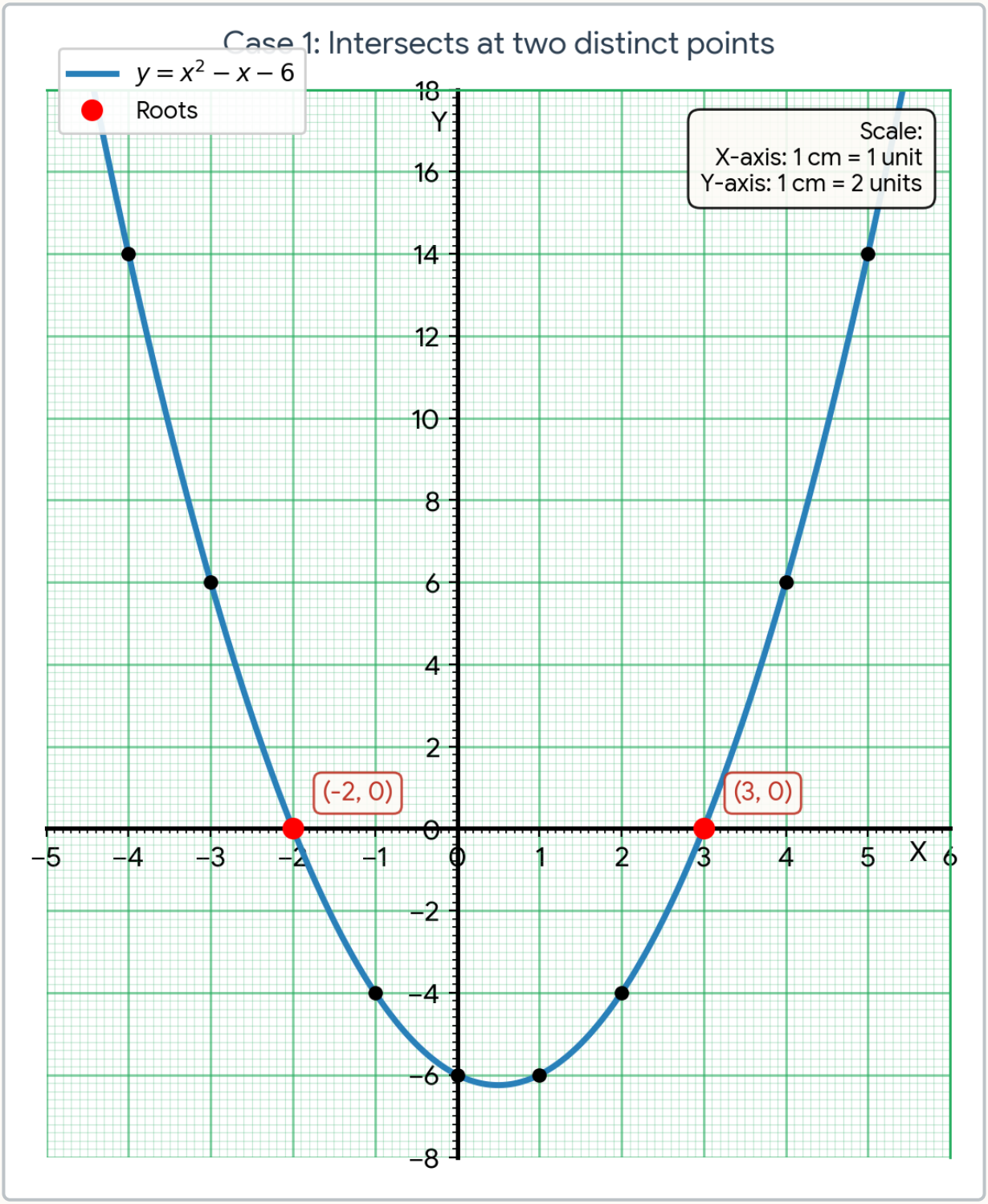
$x^2 - 3x + 2x - 6 = 0$

$x(x - 3) + 2(x - 3) = 0$

$(x + 2)(x - 3) = 0$

Therefore, $x = -2$ or $x = 3$

x	-3	-2	-1	0	1	2	3	4
x^2	9	4	1	0	1	4	9	16
-x	3	2	1	0	-1	-2	-3	-4
-6	-6	-6	-6	-6	-6	-6	-6	-6
y	6	0	-4	-6	-6	-4	0	6



Graph of y = x² - x - 6 plotted on a 1mm grid. Scale: X-axis 1cm = 1 unit, Y-axis 1cm = 2 units.

It forms a 'U' shape. It intersects the X-axis at two points. We can observe that the zeroes of this polynomial are $x = -2$ and $x = 3$. The x-coordinates of the points where the graph of this quadratic polynomial intersects the X-axis are its zeroes.

Case (ii): The graph touches the X-axis at exactly one point.

In this case, the graph touches the X-axis at exactly one point. That means the two points coincide. Thus, the points A and A' shown in Case (i) merge into a single point 'A'.

Example: Let us draw the graph for $y = x^2 - 6x + 9$.

$p(x) = x^2 - 6x + 9$

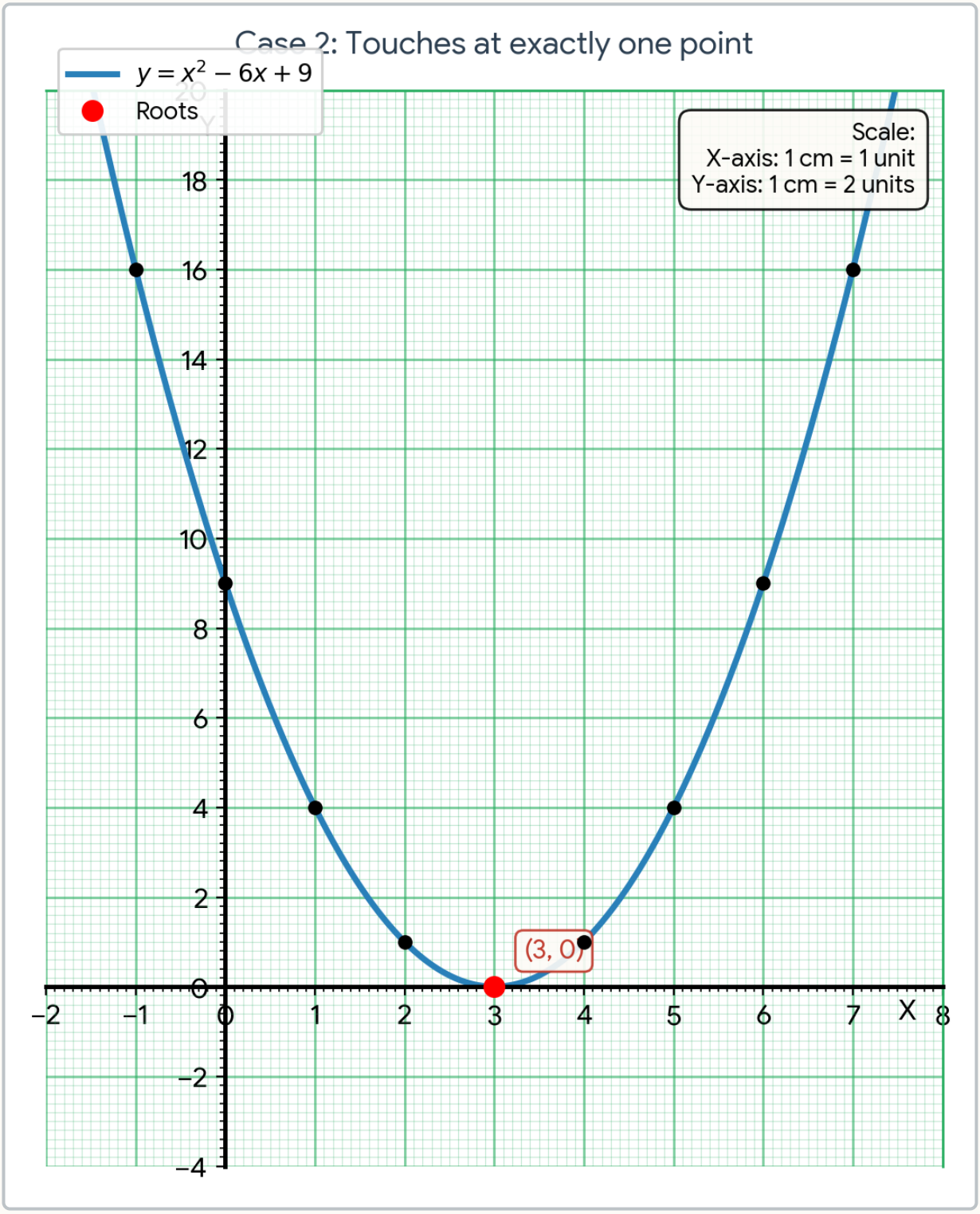
$x^2 - 3x - 3x + 9 = 0$

$x(x - 3) - 3(x - 3) = 0$

$(x - 3)(x - 3) = 0$

Therefore, $x = 3$ and $x = 3$

x	-1	0	1	2	3	4	5
x^2	1	0	1	4	9	16	25
-6x	6	0	-6	-12	-18	-24	-30
9	9	9	9	9	9	9	9
y	16	9	4	1	0	1	4



Graph of y = x² - 6x + 9 plotted on a 1mm grid. Scale: X-axis 1cm = 1 unit, Y-axis 1cm = 2 units.

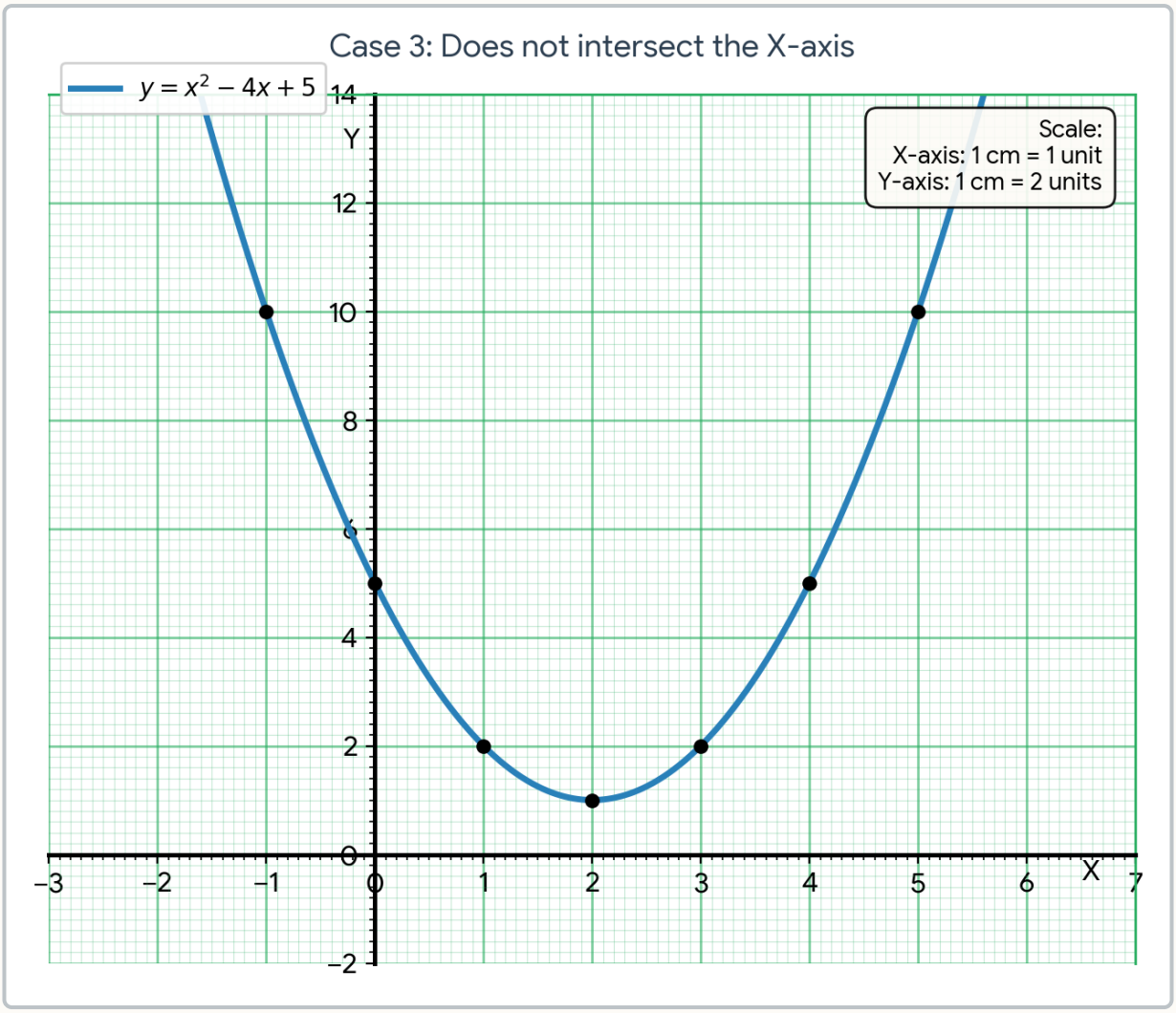
The graph forms a 'U' shape and touches the X-axis at only one point. The zero of this polynomial is $x = 3$.

Case (iii): The graph does not intersect the X-axis at any point.

In this case, the graph is either completely above or completely below the X-axis. It does not intersect the X-axis at any point.

Example: Let us find the values of x and y for the quadratic polynomial $x^2 - 4x + 5$.

x	-1	0	1	2	3	4	5
x^2	1	0	1	4	9	16	25
-4x	4	0	-4	-8	-12	-16	-20
5	5	5	5	5	5	5	5
y	10	5	2	1	2	5	10



Graph of y = x² - 4x + 5 plotted on a 1mm grid. Scale: X-axis 1cm = 1 unit, Y-axis 1cm = 2 units.

It forms a 'U' shape. It does not intersect the X-axis at any point. Therefore, this quadratic polynomial has no real zeroes.

4. Result

- In the first case, the graph intersected the X-axis at two points, so it has two zeroes.
- In the second case, it touched the X-axis at only one point, so it has exactly one zero.
- In the third case, it did not intersect the X-axis at any point, so it has no real zeroes.
- Overall, it can be concluded that a quadratic polynomial can have a maximum of two zeroes.

5. Conclusion

The zeroes of any quadratic polynomial can be found and verified by drawing its graph.

6. Reference Books Used

- 10th Class Mathematics Textbook

7. Acknowledgements

I express my heartfelt gratitude to my teacher and my group members who helped me in completing this project.